03-02

Thomson 模型的电极化

求电极化率 α $(\vec{p} = \alpha \vec{E})$

$-\frac{3Q}{4\pi a^3}$

$r=a$

球形电子云对离中心 d 处原子核 (Q) 作用力：

$$\vec{F} = \frac{Q}{4\pi\varepsilon_0}\int_V \rho(r)\cdot\frac{dV}{r^2}\hat{r} \qquad V = \{(x,y,z)\,|\,x^2+y^2+z^2 \le a^2\}$$

$$= -\frac{3Q^2}{16\pi^2\varepsilon_0 a^3}\int_V \frac{\vec{r}}{r^3}\,dV$$

$$= -k\int_V \frac{(x-d,\,y,\,z)}{[(x-d)^2+y^2+z^2]^{3/2}}\,dx\,dy\,dz$$

分别计算不同分量。（由对称性，y、z 分量均为0）

① $$\int_V \frac{x-d}{[(x-d)^2+y^2+z^2]^{3/2}}\,dx\,dy\,dz$$

$$= \int_{-a}^{a} dx \iint_{D(x)} \frac{x-d}{[(x-d)^2+y^2+z^2]^{3/2}}\,dy\,dz \qquad \text{其中 } D(x) = \{(y,z)\,|\,y^2+z^2 \le a^2-x^2\}$$

$$= \int_{-a}^{a} dx \iint_{D(x)} \frac{x-d}{(x-d)^2+\rho^2}\,\rho\,d\rho\,d\theta \qquad (\rho = \sqrt{y^2+z^2})$$

$$= \int_{-a}^{a} dx \cdot 2\pi \int_0^{\sqrt{a^2-x^2}} \frac{\frac{1}{2}(x-d)\,d(\rho^2)}{(x-d)^2+\rho^2}$$

$$= \pi \int_{-a}^{a} (x-d)\ln\left[\rho^2+(x-d)^2\right]\Big|_{\rho=0}^{\rho=\sqrt{a^2-x^2}}\,dx$$

$$= \pi \int_{-a}^{a} (x-d)\ln\frac{a^2+d^2-2dx}{(x-d)^2}\,dx$$

$$\int_{-a}^{a} x\ln(a^2+d^2-2dx)\,dx$$

$t = a^2+d^2-2dx \Rightarrow x = \frac{1}{2d}(a^2+d^2-t)$

$$= \frac{1}{2}\int_{(a+d)^2}^{(a-d)^2} \frac{1}{2d}(a^2+d^2-t)\ln t\left(\frac{-1}{2d}\right)dt$$

$$= \frac{1}{8d^2}\int_{(a-d)^2}^{(a+d)^2} (a^2+d^2-t)\ln t\,dt \qquad \int \ln t\,dt = t\ln t - t$$

$$\int t\ln t\,dt = \frac{1}{2}t^2\ln t - \frac{1}{4}t^2$$

$$= \frac{1}{8d^2}\left[(a^2+d^2)t(\ln t-1) - \frac{1}{2}t^2\left(\ln t-\frac{1}{2}\right)\right]_{(a-d)^2}^{(a+d)^2}$$

$$= \frac{1}{8d^2}\left\{(a^2+d^2)(a+d)^2\left[2\ln(a+d)-1\right] - \frac{1}{2}(a+d)^4\left[2\ln(a+d)-\frac{1}{2}\right]\right\}$$

$$- (a^2 + d^2)(a^2 - a^2)\left[2\ln(a-d) - 1\right] - \frac{1}{2}(a-d)^4\left[2\ln(a-d) - \frac{3}{4}\right] \quad ??$$

封闭球壳定理

$$Q_{eff} = -Q\cdot\left(\frac{a}{d}\right)^3$$

$$F = -\frac{Q^2}{4\pi\varepsilon_0}\cdot\left(\frac{a}{d}\right)^3\cdot\frac{1}{d^2} = -\frac{Q^2\alpha}{4\pi\varepsilon_0 a^2}$$

$$\rightarrow E = \frac{-F}{Q} = \frac{Q\alpha}{4\pi\varepsilon_0 a^2} = \frac{P}{4\pi\varepsilon_0 a^3} \Rightarrow \alpha = 4\pi\varepsilon_0 a^3$$

03-03

极化的三种物理表述
$$\begin{cases} \vec{P} \\ Q' \ (\sigma' / \rho') \\ \vec{E} = \vec{E_0} + \vec{E'} \end{cases}$$

$$\sigma' = \vec{P}\cdot\vec{n} \qquad \rho' = \nabla\cdot\vec{P}$$

03-05

极化电荷与自由电荷之关系

$$\oiint \vec{P}\cdot d\vec{S} = \varepsilon_0\chi \oiint \vec{E}\cdot d\vec{S}$$

$$\hookrightarrow -\Sigma Q' = \chi\ \Sigma(Q_0 + Q')$$

$$\Rightarrow \Sigma Q' = -\frac{\chi}{1+\chi}\Sigma Q_0$$

电场线的折射（假定界面无自由电荷）

环路定理： $E_{1t} = E_{2t}$

高斯定理（对 $\vec{D}$）： $D_{1n} = D_{2n}$

$\hookrightarrow \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}$

< V ^ >

例 2.2.24.

$$\oint D\cdot dS = 2D A = 2Ar\rho \quad \to \quad D=\rho r$$

$$E = \frac{D}{\varepsilon} = \frac{1}{\varepsilon}\rho r$$

$$P = D - \varepsilon_0 E = (\varepsilon-\varepsilon_0)\cdot\frac{1}{\varepsilon}\rho r = (1-\frac{\varepsilon_0}{\varepsilon})\rho r$$

$$\rho' = -\frac{\chi}{1+\chi}\rho = -\frac{\frac{\varepsilon-\varepsilon_0}{\varepsilon_0}}{1+\frac{\varepsilon-\varepsilon_0}{\varepsilon_0}}\rho = -\frac{\frac{\varepsilon}{\varepsilon_0}-1}{\frac{\varepsilon}{\varepsilon_0}}\rho = -\frac{\varepsilon-\varepsilon_0}{\varepsilon}\rho$$

$$\sigma' = \vec{P}\cdot\vec{n}$$

例 2.2.17

$$\sigma = \vec{P}\cdot\vec{n} = P\cos\theta$$

$$dE' = \frac{\sigma' 2\pi R^2 \sin\theta\, d\theta}{4\pi\varepsilon_0 R^2}\cos\theta$$

$$E = \int_0^{\pi} dE'$$

$$= \frac{P}{2\varepsilon_0}\int_0^{\pi}\sin\theta\cos^2\theta\, d\theta = \frac{P}{2\varepsilon_0}\int_{-1}^{1}u^2\cdot du = \frac{P}{3\varepsilon_0}$$

03.08

电介质的电场能问题

(1) $W = \frac{Q^2}{2C} = \frac{1}{2}CU^2 = \frac{1}{2}\varepsilon_r C_0 U^2 = \varepsilon_r W_0$

(2) $W = \int \frac{1}{2}\vec{D}\cdot\vec{E}\, dV$

证明：

在电介质中引入微量自由电荷 $\Delta\rho_0(\vec{r})$

$$\Delta W = \int_V \Delta\rho_0(\vec{r})\, u\, dV$$

$$= \int_V \nabla\cdot(\Delta\vec{D})\, u\, dV \qquad \nabla\cdot[(\Delta\vec{D})u] = \nabla\cdot(\Delta\vec{D})\, u + \Delta\vec{D}\cdot\nabla u$$

$$= \int_V \{\nabla\cdot[(\Delta\vec{D})u] - \Delta\vec{D}\cdot\nabla u\}\, dV$$

$$= \int_V [\nabla\cdot(\Delta\vec{D}\cdot u) + \Delta\vec{D}\cdot\vec{E}]\, dV$$

$$= \int_{\partial V}(\Delta\vec{D}\cdot u)\cdot d\vec{S} + \int_V \Delta\vec{D}\cdot\vec{E}\, dV$$

$u|_{\partial V} = 0$

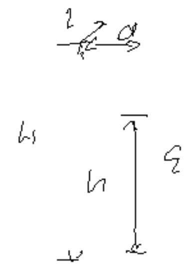
$= \varepsilon \int_V \vec{E}\cdot \delta\vec{E}$

$= \frac{1}{2}\varepsilon \int_V \delta(\vec{E}^2) = \frac{1}{2}\int_V \delta(\vec{D}\cdot\vec{E})\, dv$

$= \frac{1}{2}\,\delta\left[\int_V (\vec{D}\cdot\vec{E})\, dv\right]$

$W = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}\, dv$

p. 118 例



$C = C_1 + C_2 = \frac{\varepsilon_0 (H-h) l}{d} + \frac{\varepsilon h l}{d}$

$= \frac{l}{d}\left[\varepsilon_0 H + (\varepsilon - \varepsilon_0) h\right]$

$W = \frac{1}{2} C U^2 = \frac{l U^2 \varepsilon_0}{2d}\left[H + (\varepsilon_r - 1) h\right]$

① $dW = f\, dh = \rho g h\, d\, l\, dh$

$\Rightarrow h = \frac{l U^2 \varepsilon_0}{2\rho g d^2 l}(\varepsilon_r - 1) = \frac{\varepsilon_0 \chi U^2}{2\rho g d^2}$

② $dW = d(QU) - dA$ 一样吗？

例 2.2.1

$q' = -\frac{\chi}{1+\chi} q$

$E = \frac{q+q'}{4\pi\varepsilon_0 r^2} = \frac{q}{4\pi\varepsilon_0 r^2}\left(1 - \frac{\chi}{1+\chi}\right) = \frac{q}{4\pi\varepsilon r^2}$

例 2.2.31 ~

例 2.2.35

$\frac{4d}{R} = \frac{\sigma^2}{2\varepsilon_0} \Rightarrow d = \frac{\sigma^2}{8\varepsilon_0} R = \frac{q^2}{96\pi^2\varepsilon_0 R^3}$

例 2.3.11

$$\varepsilon = \varepsilon_1 + \frac{\varepsilon_2-\varepsilon_1}{d} t$$

$$U = \int E \cdot dl = \int_0^d \boxed{\frac{\sigma}{\varepsilon}} dt = \frac{Q}{S} d \int_0^d \frac{dt}{\varepsilon_1 d + (\varepsilon_2-\varepsilon_1)t}$$

$$= \frac{Qd}{S(\varepsilon_2-\varepsilon_1)} \ln[(\varepsilon_2-\varepsilon_1)t + \varepsilon_1 d] \Big|_0^d = \frac{Qd}{S(\varepsilon_2-\varepsilon_1)} \ln(\frac{\varepsilon_2}{\varepsilon_1})$$

$$C = \frac{Q}{U} = \frac{(\varepsilon_2-\varepsilon_1)S}{\ln(\frac{\varepsilon_2}{\varepsilon_1}) d}$$

$$D = \varepsilon E = \varepsilon \cdot \frac{\sigma}{\varepsilon} = \frac{Q}{S}$$

$$P = D - \varepsilon_0 E = \frac{Q}{S}(1 - \frac{\varepsilon_0}{\varepsilon})$$

$$\rho' = -\nabla \cdot P = \frac{Q\varepsilon_0}{S} \cdot \frac{d}{dt}(\frac{1}{\varepsilon}) = -\frac{Q\varepsilon_0}{S} \frac{1}{\varepsilon^2} \cdot \frac{\varepsilon_2-\varepsilon_1}{d} = -\frac{Q\varepsilon_0(\varepsilon_2-\varepsilon_1)}{Sd(\varepsilon_1 + \varepsilon_2-\varepsilon_1/d \cdot t)}$$

$$\sigma_1' = -P\Big|_{t=0} = \frac{Q}{S}(1 - \frac{\varepsilon_0}{\varepsilon_1})$$

$$\sigma_2' = P\Big|_{t=d} = \frac{Q}{S}(1 - \frac{\varepsilon_0}{\varepsilon_2})$$